(Please write your Enrollment Number) Enrollment No._

End-Term Examination
(CBCS)(SUBJECTIVE TYPE) (OffLine)
Course Name: B.Tech, Semester: 1
((May, 2025)

Subject Code: BAS 104	Subject: ENVIRONMENTAL SCIENCES
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1			(2.5*8=20)	CO Mapping
	a) Write a short note on common approaches towards water conservation practiced in India.			CO 1
	b) Define Chemical Oxygen Demand? How is it different from BOD? How will you determine COD?			CO 1
	c) Define carbon credits. Write a short note on international emission trading.			CO 2
	d) What is In situ biodegradation? Write its two approaches for treatment of hazardous waste.			CO 2
	e) Define Bioplastics. Explain two methods involved in the manufacture of Bioplastics.			CO 3
	f) Give the importance of the use of Photostabilizers and Photosensitisers in polymers with suitable examples			CO 3
	g) What is Ocean Thermal Energy Conversion (OTEC)? Give its merits.			CO 4
	h) List some positive and negative impacts of developing a project in an area.			CO 4
	UNIT I			CO Mapping
Q2	(i)	50 ml of a sample of water required 5 ml of N/50 H_2SO_4 using methyl orange as indicator but did not give any coloration with phenolphthalein. What type of alkanity is present? Express the same in ppm.	(5,5)	CO 1
	(ii)	Discuss the various steps involved in the primary treatment of waster water.		
Q3	(i)	(a) The total hardness of a sample of water was determined by titrating the sample with standard EDTA solution. From the following results calculate the total hardness of water sample and express the hardness of water in ppm. 100 mL of water required 10.2 mL of centimolar EDTA. The indicator used is EBT. (b) why EDTA titrations are performed in buffer of pH 8-10?	(5,5)	CO 1
	(ii)	(a) Explain 'Trickling filters' of waste water treatment (b) Discuss three methods for the removal of dissolved pollutants from the waste water.		

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		UNIT II		**CO Mapping**
Q4	(i)	Write five principles of Green Chemistry. Explain them with proper chemical reactions, if any.	**(5,5)**	**CO 2**
	(ii)	What are sanitary landfills? Discuss the advantages and problems of sanitary landfills.		
Q5	(i)	What is acid rain? Explain the chemistry behind the formation of acid rain.	**(5,5)**	**CO 2**
	(ii)	What is the % atom economy of the desired product in the following reaction (a) $Fe_2O_3 + 3CO \longrightarrow 2Fe$ (Desired Product) $+ 3CO_2$ (b) Calculate the atom economy for the preparation of ammonia from the thermal decomposition of ammonium sulfate $(NH_4)_2 SO_4 \longrightarrow H_2SO_4 + 2NH_3$ (Desired Product)		
		UNIT III		**CO Mapping**
Q6	(i)	Give the mechanism of degradation of photo biodegradable and hydro-biodegradable polymers.	**(5,5)**	**CO 3**
	(ii)	Enumerate various forms of Mercury and describe its biochemical effects.		
Q7	(i)	Discuss various forms of chromium (Cr) with their sources and Bio chemical effects	**(5,5)**	**CO 3**
	(ii)	Equal weights of polymer molecules with molecular weights 20,000 g/mol and 200,000 g/mol are mixed. Calculate $\overline{Mn}$ and $\overline{Mw}$?		
		UNIT IV		**CO Mapping**
Q8	(i) Discuss five important components of Environmental Management Systems? Explain. (ii) write a short note on (a) Renewable and non-renewable sources of energy (b) Concept of sustainable development		**(5,5)**	**CO 4**
Q9	(i)	What is Power alcohol? Explain the steps/processes involved in its formation.	**(5,5)**	**CO 4**
	(ii)	What are the challenges associated with Green hydrogen energy		

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